EE3402- Linear Integrated Circuits

Mini-Project List

Final Demonstrations is scheduled tentatively on May 17, 2025

Team of 3 to 4 can present design and demonstrate any one of the projects listed (but it is not limited to following).

Total Number of Teams is limited to 12 to 15.

Expected Outcome:

Students has to explain during the hardware demonstration:

1. Design of the circuits
2. Components selection
3. Working
4. Advantages implemented circuit over the other similar circuits.
5. Submit a report 10-15 pages.

Op-Amp Based Projects

1. Precision Rectifier using Op-Amp
2. Op-Amp Based Temperature Sensor with Analog Display
3. Audio Amplifier using 741 Op-Amp
4. Op-Amp Based Light Detector with Variable Gain
5. Op-Amp Based Function Generator (Sine, Square, Triangle)
6. Analog PID Controller using Op-Amps
7. Voltage Level Detector using Op-Amp Comparator
8. Op-Amp Based Active Band Pass Filter
9. Analog Multiplier Circuit using Op-Amps
10. Automatic Gain Control (AGC) Circuit using Op-Amp

555 Timer Based Projects

11. Buck Converter Control using 555 Timer
12. Boost Converter Control using 555 Timer
13. Digital Dice using 555 Timer and Logic Gates
14. Speed Control of DC Motor using 555 Timer PWM
15. Traffic Light Controller using 555 and Decade Counter
16. Water Level Alarm using 555 Timer

17. Electronic Eye (LDR based Security Alarm)
18. Burglar Alarm System using 555 Timer
19. Rain Alarm using 555 Timer
20. Touch-based Switch using 555 Timer

Combined Op-Amp and 555 Timer Based Projects

21. Smart Light Controller using LDR, Op-Amp, and 555 Timer
22. Tone Generator with Frequency Modulation using Op-Amp and 555 Timer
23. Pulse Width Modulation Control using Op-Amp Comparator and 555
24. Heart Beat Monitor using Op-Amp and 555 Timer
25. Fire Alarm System using Thermistor, Op-Amp, and 555 Timer